
SOFTWARE REQUIREMENTS SPECIFICATION

For

**Web-Based Patient Scheduling and
Program Management System for
Barangay Health Center**

Prepared by:

Celestial ART

Angala, Regin

Rosal, Angel

Tabunyag, Jonalyn

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1. Introduction

In the present time, people can no longer imagine their lives without technology. The rapid growth of the technology sector has led to the development of numerous innovations that have transformed the way people work, communicate, and live. Among these, the computer stands out as one of the most significant tools, as it supports individuals in completing office tasks, conducting business transactions, and providing various forms of entertainment. Computers, along with modern technologies, have made manual processes faster, easier, and more efficient.

As society continues to progress, individuals and organizations increasingly rely on technology to enhance their daily activities. One particular sector that benefits greatly from technological advancements is healthcare. Health centers perform numerous daily tasks, from managing patient records to scheduling appointments, which are often time-consuming when done manually.

This project is design to address significant administrative and healthcare delivery challenges through a comprehensive digital solution. Barangay Health Center plays a vital role in delivering essential healthcare services to the community. However, the center faced inefficiencies caused by manual record-keeping processes and limited resources. These issues hindered effective healthcare delivery, prompting the need for modernization to optimize resource utilization and enhance access to healthcare services.

The main challenge tackled by this project was the reliance on manual systems for patient scheduling and program management. This outdated approach resulted in delays in healthcare delivery, mismanagement of resources, and challenges in implementing health programs. These inefficiencies directly affected the residents' ability to access timely and effective care. Resolving these issues was crucial not only to improve service delivery but also to align with the government's push for digitalization and modernization in healthcare.

This project focuses on the development of system design to improve the operations of health centers. One of the system's key features is its ability to automate patient appointment scheduling. This eliminates long wait times and minimizes scheduling conflicts, allowing healthcare staff to focus more on providing quality care. The system centralizes patient medical histories, ensuring that healthcare providers have access to accurate and up-to-date information. This improves the

quality of care and enables better decision-making. Centralized records also allow for seamless program coordination, which is vital for implementing health initiatives in Barangay. With its program management features, the system optimizes resource allocation and ensures the timely execution of healthcare initiatives. From immunization to family planning programs, the system simplifies coordination and enhances outcomes. This aligns with the government's digitalization initiatives aimed at improving public healthcare services.

1.1 Purpose

The Web-Based Patient Scheduling and Program Management System for Barangay Health Center is design to automate key processes at Barangay Health Center. It will streamline patient appointment scheduling, track medical histories, and facilitate program management. Additionally, the system will provide tools for generating automated reports and reminders, which enhanced administrative workflows and reduced manual workload. The design prioritized user-friendliness, security, and scalability to ensure that the system could adapt to future healthcare needs. The solution was tailored to the health center's unique requirements, ensuring its practicality and long-term relevance.

1.2 Intended Audience

- Patients/ Residents: Book appointments, view medical history, receive notifications, and participate in health programs.
- Health Workers:
 - Doctors: Consult with patients, update medical records, and manage appointments.
 - Nurses: Assist in patient care, record vitals, and support clinical procedures.
 - Midwives: Manage prenatal/postnatal care and family planning services.
- Barangay/Community Health Personnel: Manage user credentials and generate reports.

1.3 Product Scope

The system is a comprehensive web-based platform that includes:

Core Features

1. Role-Based Dashboards

- **Patient Dashboard:** Appointment booking, medical history, notifications, profile management.
- **Health Worker Dashboard** (Doctor/Nurse/Midwife): Patient consultations, appointment management, medical notes, service tracking.
- **Barangay Health Personnel Dashboard:** User credential management, master list processing, analytics, report generation.

2. Appointment Management

- Online booking with date/time selection.
- Appointment status tracking (Scheduled, In Progress, Completed, Canceled).
- Automated reminders and notifications.
- Rescheduling and cancellation by patients or health workers.

3. Patient & Medical Records

- Secure storage of patient profiles, medical history, and appointment records.
- Child record management for family healthcare.
- Access control based on user roles.

4. Health Program Management

- Program scheduling and participant tracking.
- Resource allocation and progress monitoring.
- Integration with appointment and patient data.

5. Notification System

- Browser notifications for real-time alerts.
- Email reminders for appointments and announcements.
- In-app notification center with mark-as-read and delete functions.

6. Reporting & Analytics

- Real-time statistics on appointments, patients, and programs.
- Chart visualizations (pie, line, bar, radar) for data analysis.

- Exportable reports in JSON format.

7. User & Credential Management

- Bulk user creation via masterlist upload (CSV/TXT).
- Individual user creation with validation.
- Role assignment (Patient, Doctor, Nurse, Midwife, BHW).

1.4 Definitions, Acronyms, and Abbreviations

1.4.1 Definitions

- **Web-Based System** – A software application accessible via web browsers, hosted on a server, requiring no local installation.
- **Patient Scheduling** – Digital process of booking, managing, and tracking patient appointments with healthcare providers.
- **Program Management** – The planning, coordination, and execution of healthcare-related activities or initiatives such as immunization and maternal care programs within the community.
- **Dashboard** – Role-specific interface displaying key information, actions, and navigation for system users.
- **Master List Processing** - Bulk import of user data from structured files (CSV/TXT) for credential generation.
- **Healthcare Provider** – Refers to doctors, nurses, and midwives who deliver medical and healthcare services to the community.
- **Barangay Health Center** – A community-level healthcare facility that provides primary medical and preventive services to residents within a barangay.

1.4.2 Acronyms and Abbreviations

- **BHC** – Barangay Health Center
- **BHW** – Barangay Health Worker
- **DB** – Database
- **EHRS** – Electronic Health Record System
- **UI** – User Interface
- **UX** – User Experience
- **DOH** – Department of Health

2. Overall Description

2.1 User Characteristics

- **Patients:** Basic computer/mobile literacy; minimal training required.
- **Doctors/Nurses:** Familiar with medical record handling; may require short orientation on digital usage.
- **Midwives:** They act as community front-liners, providing health education on family planning and immunization, and referring high-risk cases to hospitals.
- **System Admin:** Develops and manages server security, backup and recovery plans.
- **Barangay Health Personnel:** The one who generate reports, create and manage residents' login credentials

2.2 Constraints

The system development and implementation are subject to the following limitations:

- **Internet Dependency:** Requires stable internet for real-time functionality.
- **Budget** – Financial constraints may limit the procurement of advanced technologies or additional system features beyond the essential requirements.
- **Hardware Limitations:** May run on older devices; responsive design required.
- **Digital Literacy:** Some users may require additional support or training.
- **Data Privacy and Compliance** – The system must adhere to data privacy regulations (e.g., Data Privacy Act of 2012 in the Philippines), which may restrict certain data-handling procedures.

2.3 Assumptions and Dependencies

- Users (patients, doctors, nurses, midwives, and Barangay Health Workers) will have access to mobile devices or computers with internet connectivity to interact with the system.

- The Barangay Health Center will provide the necessary infrastructure (internet connection, computers, and power supply).
- Users will cooperate in adopting the system, including attending basic training sessions.
- Patients will have valid and accessible email addresses for receiving appointment reminders, updates, and follow-up notifications.
- Patient data provided during registration will be accurate and updated regularly by healthcare personnel.

3. Requirements Specifications

3.1 Functional Requirements.

- **Login**
 - Support secure login with credentials (username and password).
 - Provide role-based access (patients, health workers, and admin).
- **Authentication & Security**
 - Encrypt passwords before storing them in the database.
 - Implement session management to prevent unauthorized access.
- **Profile Management**
 - Users (patients & health workers) can view and update personal information.
 - Health workers can update professional details (specialization, role).
 - Admin can update or delete user accounts.
- **Admin Dashboard**
 - Manage all registered users (view, add, edit, delete).
 - Oversee system logs for monitoring and auditing.
- **System Feedback**
 - Display confirmation/error messages during signup, login, and account updates.

- Provide validation for incorrect input (e.g., empty fields, invalid email format).

3.2 Non-Functional Requirements

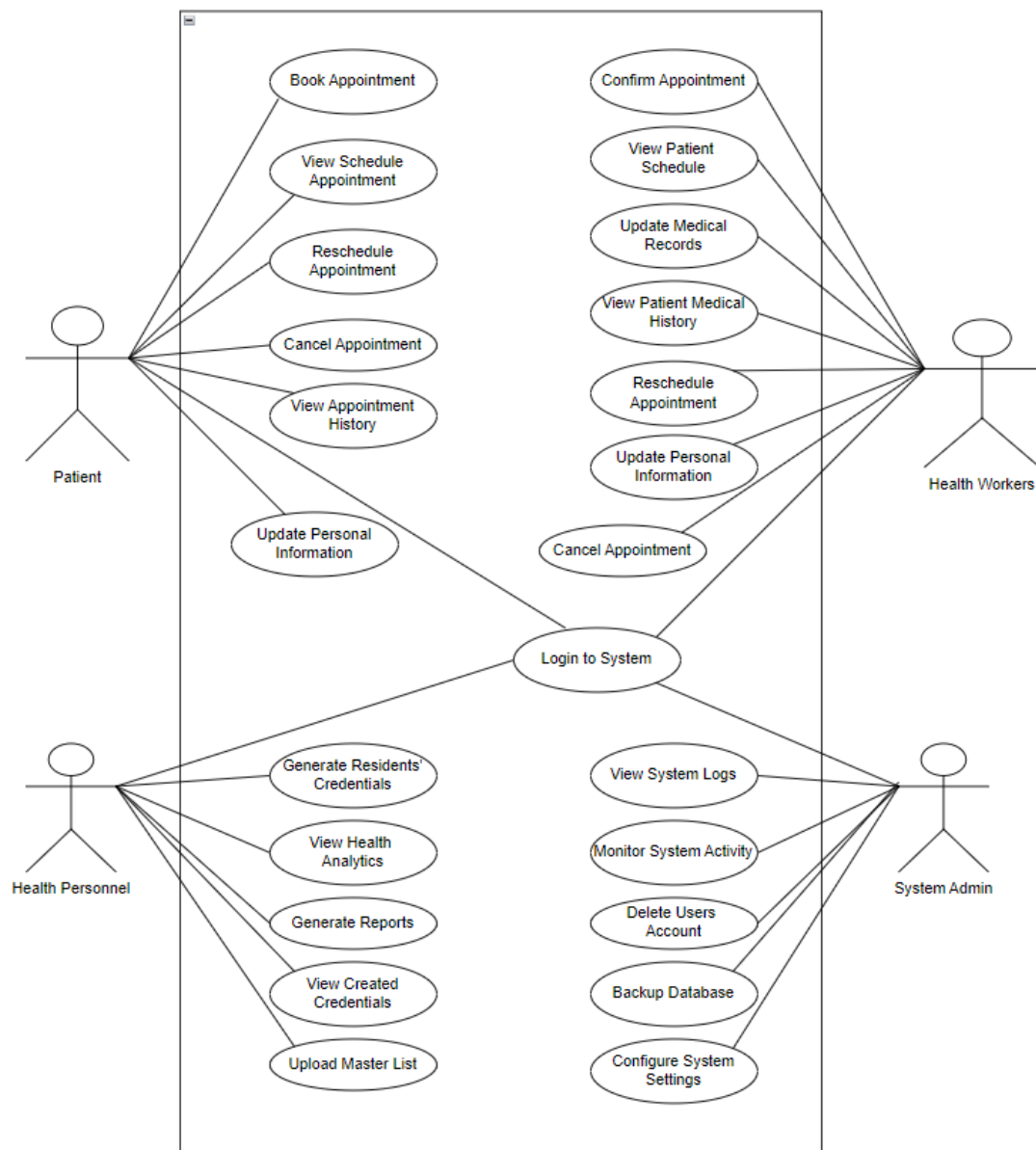
- **Performance:** Page load < 3 seconds, appointment booking < 5 seconds, real-time updates.
- **Security:** Password encryption, HTTPS, session timeout, role-based access, data privacy compliance.
- **Usability:** Intuitive UI, mobile-responsive design, clear navigation, minimal training required.
- **Reliability:** 95% uptime, error handling, data backup and recovery mechanisms.
- **Scalability:** Support for 500+ concurrent users, expandable database, modular architecture.

3.3 External Interface Requirements

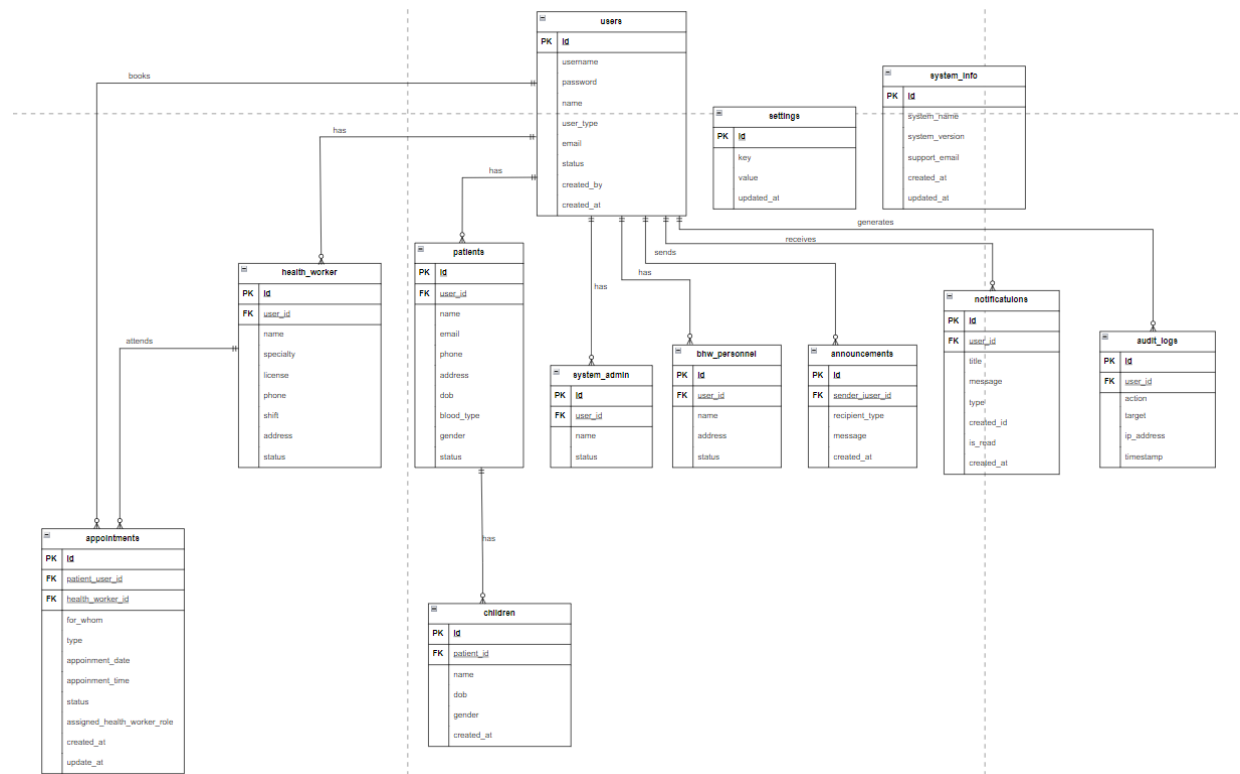
- **User Interfaces:** Responsive web design (Bootstrap/CSS), role-based dashboards, accessible via Chrome, Firefox, Edge, Safari.
- **Hardware Interfaces:** Compatible with desktops, laptops, tablets, and smartphones.
- **Software Interfaces:** PHP backend, MySQL database, RESTful APIs, Chart.js for visualizations.
- **Communication Interfaces:** Browser Notification API for real-time alerts.

3.4 System Model

Use Case Diagram



3.5 Database Design



3.6 Implementation

The development of the *Web-Based Patient Scheduling and Program Management System* was carried out using a structured and modular approach to ensure maintainability, scalability, and ease of deployment. The system was implemented in a web-based environment to allow accessibility through standard web browsers on both desktop and mobile devices.

Development Environment

The system was developed using a Windows-based environment with Visual Studio Code as the primary integrated development environment (IDE). A local web server was configured using XAMPP, which provided Apache, MySQL, and PHP for development and testing. The system was tested locally before deployment to a live server environment.

- **Operating System:** Windows 10 / 11
- **IDE:** Visual Studio Code
- **Local Server:** XAMPP (Apache, MySQL, PHP)

- **Web Browser for Testing:** Google Chrome

Programming Languages

The project utilized a combination of **front-end** and **back-end** programming languages to ensure interactive and secure web functionality.

- **HTML** – For creating the structure and layout of web pages.
- **CSS** – For styling and responsive design across devices.
- **JavaScript** – For client-side interactivity and dynamic content.
- **PHP** – For server-side scripting, user authentication, and database interaction.
- **(MySQL)** – For database management and query execution.

Frameworks and Libraries

To enhance development efficiency and ensure a modern user interface, several frameworks and libraries were integrated into the project:

- **jQuery** – For simplifying JavaScript operations and DOM manipulation.
- **PHPMyAdmin** – For database management and testing.

Tools and Utilities

Additional tools were used throughout the development process for version control, collaboration, and system optimization.

- **Git and GitHub** – For version control and collaborative code management.
- **Figma / Canva** – For UI/UX prototyping and design planning.
- **Postman** – For testing API endpoints and ensuring reliable communication between the client and server.
- **Google Fonts & Icons** – For enhancing the visual appeal and consistency of the user interface.

3.7 Testing and Deployment

The system was initially deployed on a local XAMPP server for testing and debugging, and later configured for web hosting to enable online access by Barangay Health Center staff and patients. The deployment process included creating a secure database connection, enabling HTTPS for secure data transmission, and configuring user roles to ensure data privacy and access control.

3.7.1 Load & Stress Testing

System Context: The system is a healthcare web application that supports two main user roles: the patient and the health worker. Patients manage their health information, book appointments, and receive notifications. Health worker manage their profile information, accept and manage appointments, and send announcements or notify patients.

What we mean by:

- A user is defined as an authenticated individual accessing the portal via browser
- A transaction refers to any API call or page load action (e.g., login, load dashboard, fetch appointments).
- Load is simulated using JMeter with 1000, 2000, and 5000 concurrent users performing typical workflows: login → load dashboard → interact with APIs. Load testing was conducted locally using JMeter.

Metrics Measured:

- Response Time (seconds): How long pages take to load
- Success Rate: Whether operations complete without errors
- System stability: Does it crash or keep working under heavy load?

Testing Environment:

- Local Laptop
- Windows laptop with 8GB RAM
- Tools: JMeter

3.7.1.1 Load Test Results

System Context: The system is a healthcare web application that supports two main user roles: the patient and the health worker. Patients manage their health information, book appointments, and receive notifications. Health worker manage their profile information, accept and manage appointments, and send announcements or notify patients. Our load test simulated multiple patients accessing the system simultaneously to measure performance under normal and peak usage conditions.

Table 1: Load Test Performance

Database Load	Average Response Time	System Performance Observation
1000 records	20.60 ms	System performed optimally.
5000 records	25-28 ms	Stability is good. No noticeable impact from database growth.
10000 records	35-45 ms	First signs of memory accumulation, database connection wait time begins, but still UI is responsive.
50000 records	65-85 ms	Page loads feel sluggish and occasional timeouts for heavy operations.

The load test indicates that the system performs efficiently under light to moderate database loads. For 1,000 to 5,000 records, the system maintains low average response times (20.60–28 ms) and shows no significant performance degradation, confirming good stability. As the database size increases to 10,000 records, minor performance impacts are observed, such as memory accumulation and slightly longer database connection wait times; however, the user interface remains responsive. Under heavy load with 50,000 records, response times rise to 65–85 ms, resulting in slower page loads and occasional timeouts for intensive operations, indicating that system resources are being stressed. Overall, the system scales well for small to moderate database sizes but may require optimization (e.g., query optimization, caching, or increased server resources) to handle very large datasets efficiently.

3.7.1.2 Stress Test Result

System Context: The system is a healthcare web application that supports two main user roles: the patient and the health worker. Patients manage their health information, book appointments, and receive notifications. Health worker manage their profile information, accept and manage appointments, and send announcements or notify patients. The stress test pushed the system beyond typical load levels to identify breaking points, resource limitations, and potential bottlenecks under extreme conditions.

Table 1: Stress Test Performance

Users Tested	What Users Did	System Response	What we observed
1000 users (500 normal + 500 stress)	• Login • Book appointment • Accept and manage appointment	• Avg response time: 80 ms • Fast dashboard loading (3 ms) • APIs responded within 17–19 ms • Database operations remained under 40 ms	System remained stable under load, no crashes or service interruptions, minor 401/403 errors due to session handling and performance remained smooth and responsive.
2000 users (1000 normal + 1000 stress)	• All possible actions • Constant request hammering • Stress to locate breaking point	• Avg response time: 144 ms • Login remained stable (15–30 ms) • APIs maintained 15–17 ms • Database latency increased to 80–120 ms	System continued functioning without failure, increased database pressure observed, no hard breaking point reached, and response time degradation was gradual and predictable.
5000 users (mixed operations)	• Maximum concurrent load	• Avg response time: 377 ms	System remained operational at extreme load, no crashes or

	• Web interface testing	• Login avg: 94 ms • Dashboard loading stayed under 10 ms • APIs averaged 25 ms • Database became dominant bottleneck (250 ms)	service downtime, significant database contention observed, and minimal performance degradation, not system failure.
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The stress test demonstrates that the system performs reliably under increasing user loads, with gradual performance degradation primarily at the database layer. At 1,000 users, response times remained very low, and the system showed no stability issues. At 2,000 users, the system continued to function with slightly increased database latency, but no failure occurred. Even at 5,000 users, the system remained operational, though the database became the main bottleneck, causing moderate response time increases. Overall, the system can handle high concurrency, but database optimization is recommended to improve scalability under extreme loads.

3.7.2 User Acceptance Testing

User Acceptance Testing (UAT) is the final testing phase where the Barangay Health Center Management System (BHCMS) is evaluated by intended users, including patients and health workers (doctors, nurses, midwives), to ensure it meets functional and real-world operational requirements. Users tested key features such as appointment booking, patient management, notification systems, and profile management to confirm correct functionality, usability, and data accuracy. Successful UAT indicates that the system is ready for deployment and meets both user and business requirements.

Evaluation Dimension	Mean Score (1-5)	Percentage (%)	Verbal Interpretation
Usability	4.60	92.0%	Strong Agreement
Reliability	4.53	90.6%	Strong Agreement
Interface Design	4.50	90.0%	Strong Agreement
Overall System	4.55	91.0%	Strong Agreement

Table 3. User Acceptance Testing Result

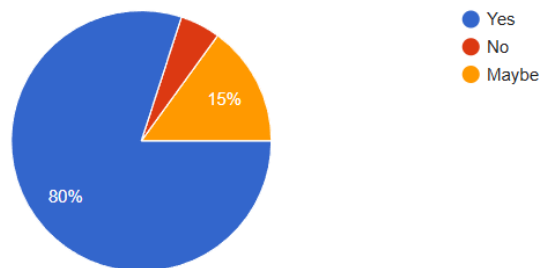
The results indicate that the system achieved an exceptionally high level of user acceptance across all evaluation dimensions. Usability obtained the highest rating at 92.0%, demonstrating that respondents found the system intuitive and straightforward to learn and operate. Interface Design followed closely at 90.0%, reflecting a consistently positive perception of the system's layout and navigational clarity. Reliability recorded a 90.6% agreement rate, indicating strong user trust in the system's stability and performance. Overall, the system achieved a 91.0% approval rating, which corresponds to a Strong Agreement interpretation across all user groups, including both patients and health workers.

Pie Charts Evaluation

Would you recommend this system to other patients and/or health workers?

40 responses

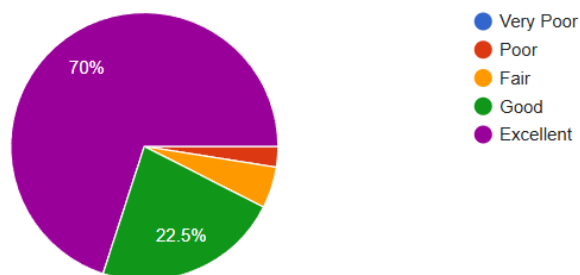
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How would you rate this system based on your experience?

40 responses

 [Copy chart](#)



Testing

The following shows the rating guide using the 5-likert scale.

Rating Guide:

5 - Strongly Agree

4 - Agree

3 - Neither Agree Nor Disagree

2 - Disagree

1 - Strongly Disagree

Usability	1	2	3	4	5
1. The system was easy to navigate and understand.					
2. The system functions worked smoothly and were well-integrated.					
3. I could learn to use the system quickly without difficulty.					
4. The interface was responsive on different devices.					
5. I felt confident using the system to perform tasks.					

Reliability	1	2	3	4	5
1. The system performed tasks consistently without errors.					
2. Pages and functions loaded properly within expected time.					
3. The system remained stable with multiple users.					
4. Data were saved and retrieved accurately.					

Reliability	1	2	3	4	5
5. The system was accessible when needed.					

Interface	1	2	3	4	5
1. The interface layout was organized and clear.					
2. Buttons and icons were consistent across pages.					
3. The interface displayed properly on computer and mobile.					
4. Text, labels, and instructions were clear and readable.					